



KAU Library Management System

30 November 2023

Presented to:
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Section: CS1

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PART I: Analysis

1. Problem Definition and Data Requirements

1.1 Problem Description

The library management system is a key element in any educational institution or community center. Management of library resources through the manual system is outdated with a variety of challenges associated with keeping track of books, creating, and updating borrowers' details as well as assuring the availability of the books. We put forward the creation of an extensive Library Management System that will assist us in addressing these concerns.

1.2 Data Requirements

1) Employee requirements

Each employee has:

- Employee ID: INT(9)
- Employee Name
- Position
- Starting Time
- Phone Number

2) Shift requirements

Each shift has:

- Starting Time: Date
- Leaving Time

3) Student requirements

Each student has:

- Student ID: INT(7)
- Student Name
- Phone Number
- Email
- Degree

4) Book requirements

Each book has:

- Book ID: INT(9)
- Book Name
- Author
- ISBN
- Edition

5) Loan requirements

Each loan has:

- Loan ID
- Loan Date
- Status

6) Genre requirements

Each genre has:

- Genre Code
- Genre Name

7) Bookshelf requirements

Each bookshelf has:

- Bookshelf ID
- Bookshelf code

8) Section requirements

Each section has:

- Section ID
- Location

9) Loan_History requirements

Each Loan_History has:

- Loan_ID
- Book_ID

10) Author_List requirements

Each Book has:

- Book_ID
- Author_Name

1.3 Business Rules

Employee:

- Employee's age between 18 years old and 60 years old.
- A section must have only one employee who manages it, and the manager cannot manage more than one section.
- Employee ID must be unique and not null.
- An employee may work in one section, but each section must have many employees.
- Each loaning process has an employee responsible for it.
- Each employee must have one address, and the address can be repeated for many employees.

Student:

- Student ID must be unique and not null.
- A student can loan at most 3 books.

Book:

- Book ID must be unique and not null.
- The book must be put on bookshelf.
- Any book can be loaned if is available.
- Each book has one section.

Bookshelf:

- A bookshelf may contain more than one book.
- Each bookshelf must be in a section.

Section:

- Section ID must be unique and not null.
- Each section has bookshelves.

Loan:

- Loan ID must be unique and not null.
- The duration for loaning a book is a maximum of two weeks.

Genre:

- Each section has at least one genre.
- Each book has one genre.

1.4 Intended Output of the System

❖ Queries:

- Books in a category.
- Students with loans.
- Books on loan.
- number of books loaned in a genre.
- number of books loaned in a section.

❖ Transactions:

- Add books.
- Remove books.
- Register new employees.
- Add a new loan.

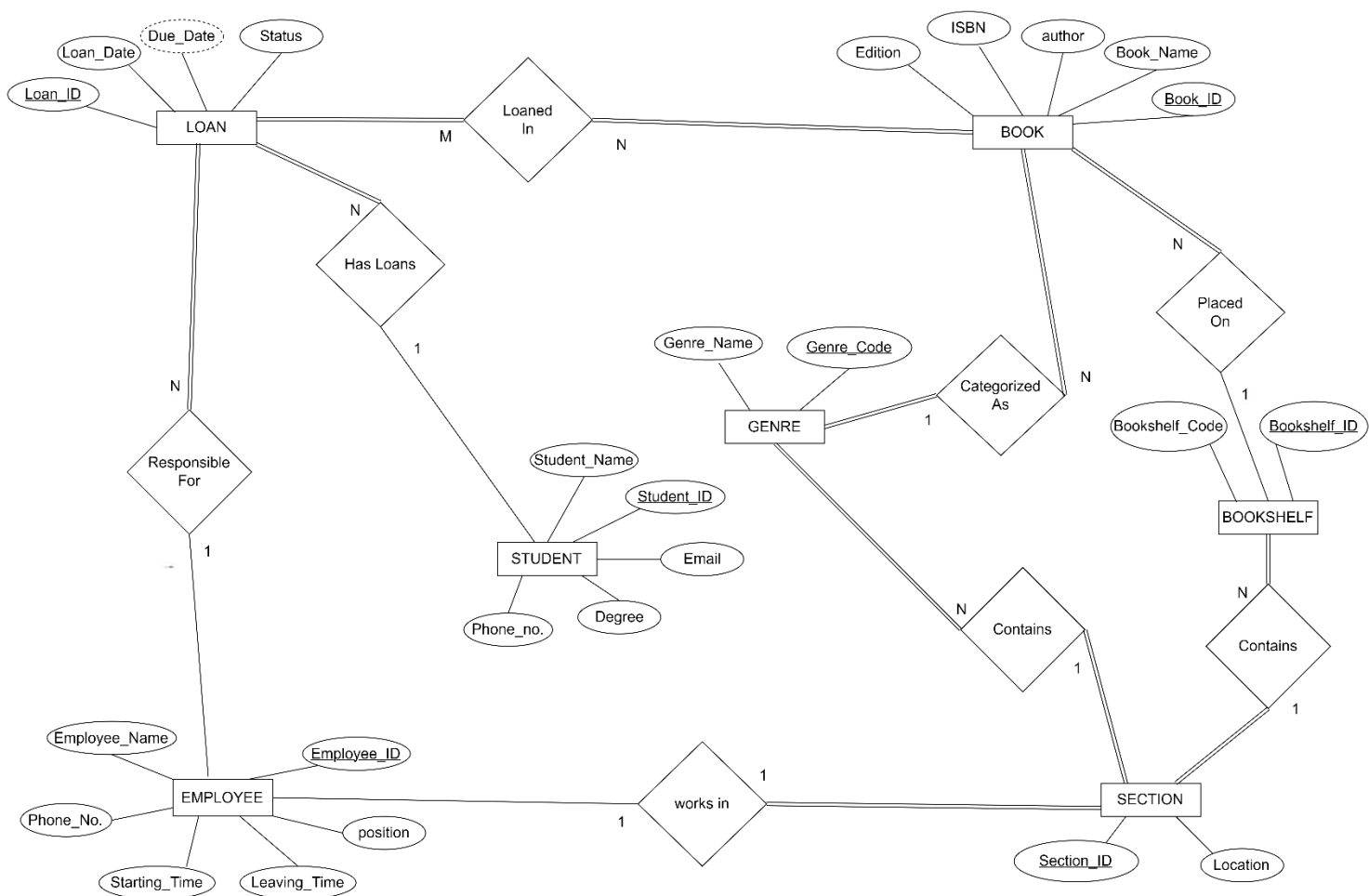
❖ Reports:

- Books in the library.
- Employees in the library.
- Most borrowed books.
- Loan history.

PART I: Analysis

2. ER Diagram Design

2.1 ER Diagram



2.2 Design of Business Rules

This subsection shows how the business rules have been translated into design decisions. Some business rules can be deployed during implementation phase only. Provide sufficient justification.

	Business Rule	Design Decisions	Justification (if any)
1	The book must be put on bookshelf, and a bookshelf may contain more than one book.	1:N binary relationship between bookshelf and book.	Many books can be stored in one bookshelf. The book entity has total participation, but the bookshelf entity has partial participation.
2	Each section has bookshelves, and each bookshelf must be in a section.	1:N binary relationship between section and bookshelf.	Partial participation from both sides.
3	A section must have only one employee who manages it, and the manager cannot manage more than one section.	1:1 binary relationship between employee and section.	Section entity has total participation with the relation (manages). However, the entity employee has partial participation.
4	A student can loan many books.	1:N binary relationship between student and loan.	A student can loan more than one book. The loan entity has total participation, but the student entity has partial participation.
5	Any book can be loaned if is available, and there is no loaning without a book.	M:N binary relationship between loan and book.	The loan entity has total participation, but the entity book has partial participation.
6	Each loaning process has an employee responsible for it, and not every employee is responsible for a loaning process.	1:N binary relationship between employee and loan.	The loan is fully participating in the relationship, while an employee has partial participation in the relationship.

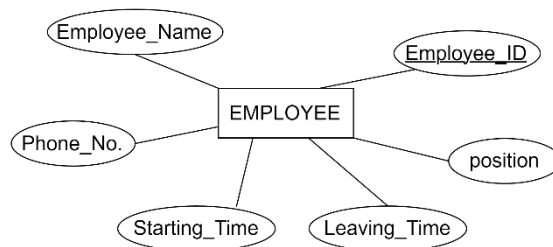
7	Each genre has more than one book.	1:N binary relationship between book and genre .	Partial participation from both sides.
8	Each section has at least one genre.	1:N binary relationship between section and genre .	Partial participation from both sides.
9	Book ID must not be null and must be unique for each one.	Primary key constraint.	Used to identify one employee from another one.
10	Employee's age must be between 18 years old and 60 years old.	A check constraint during the implementation phase is represented by this rule.	
11	A student can borrow at most 3 books.	This rule represents a check constraint in the implementation phase.	
12	The duration for loaning a book is a maximum of two weeks	A check constraint during the implementation phase is represented by this rule.	
13	Student's age must be over 15 years old.	This rule is a representation of a check constraint during implementation.	

3. ER-to-logical Schema Mapping

3.1 Mapping of Regular Entity Types

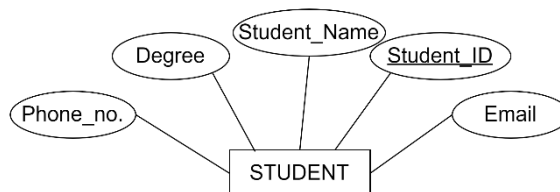
EMPLOYEE

<u>Employee_ID</u>	Employee_Name	Position	Starting_Time	Leaving_Time	Phone_No.
--------------------	---------------	----------	---------------	--------------	-----------



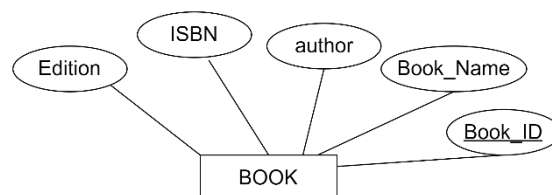
STUDENT

<u>Student_ID</u>	Student_Name	Phone_No.	Degree	Email
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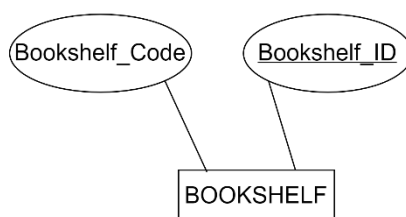
BOOK

<u>Book_ID</u>	Book_Name	ISBN	Author	Edition
----------------	-----------	------	--------	---------



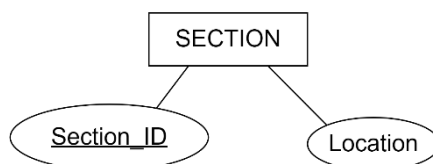
BOOKSHELF

<u>Bookshelf_ID</u>	Bookshelf_Code
---------------------	----------------



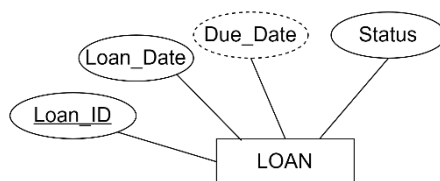
SECTION

<u>Section_ID</u>	Location
-------------------	----------



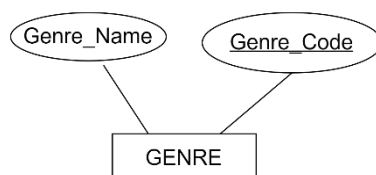
LOAN

<u>Loan_ID</u>	Loan_Date	Due_Date
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GENRE

<u>Genre_ID</u>	Genre_Name
-----------------	------------



3.2 Mapping of Weak Entity Types

There is no entity type in our database.

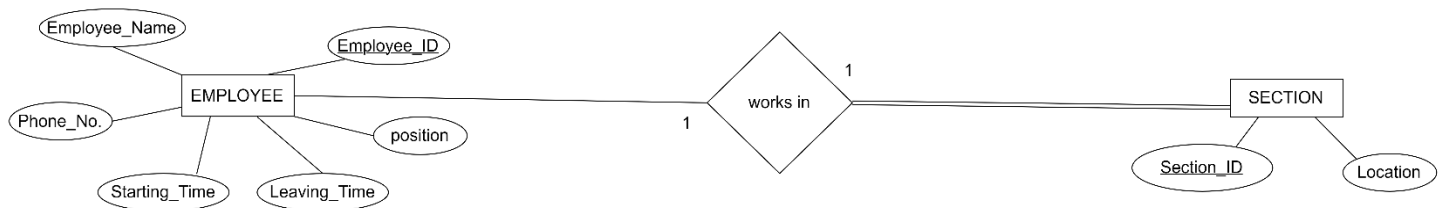
3.3 Mapping of Binary 1-1 Relationship Types

EMPLOYEE

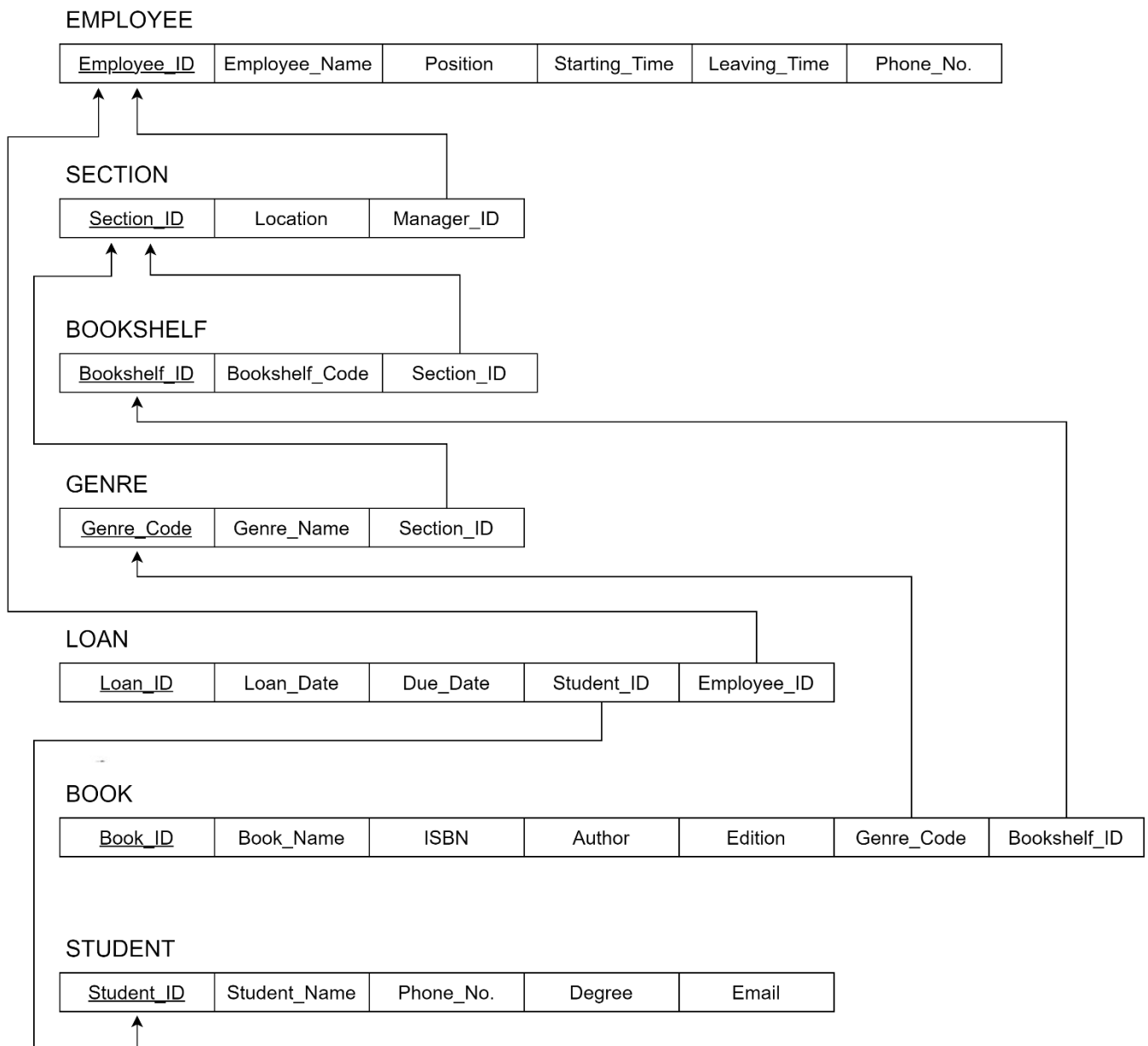
<u>Employee_ID</u>	Employee_Name	Position	Starting_Time	Leaving_Time	Phone_No.
--------------------	---------------	----------	---------------	--------------	-----------

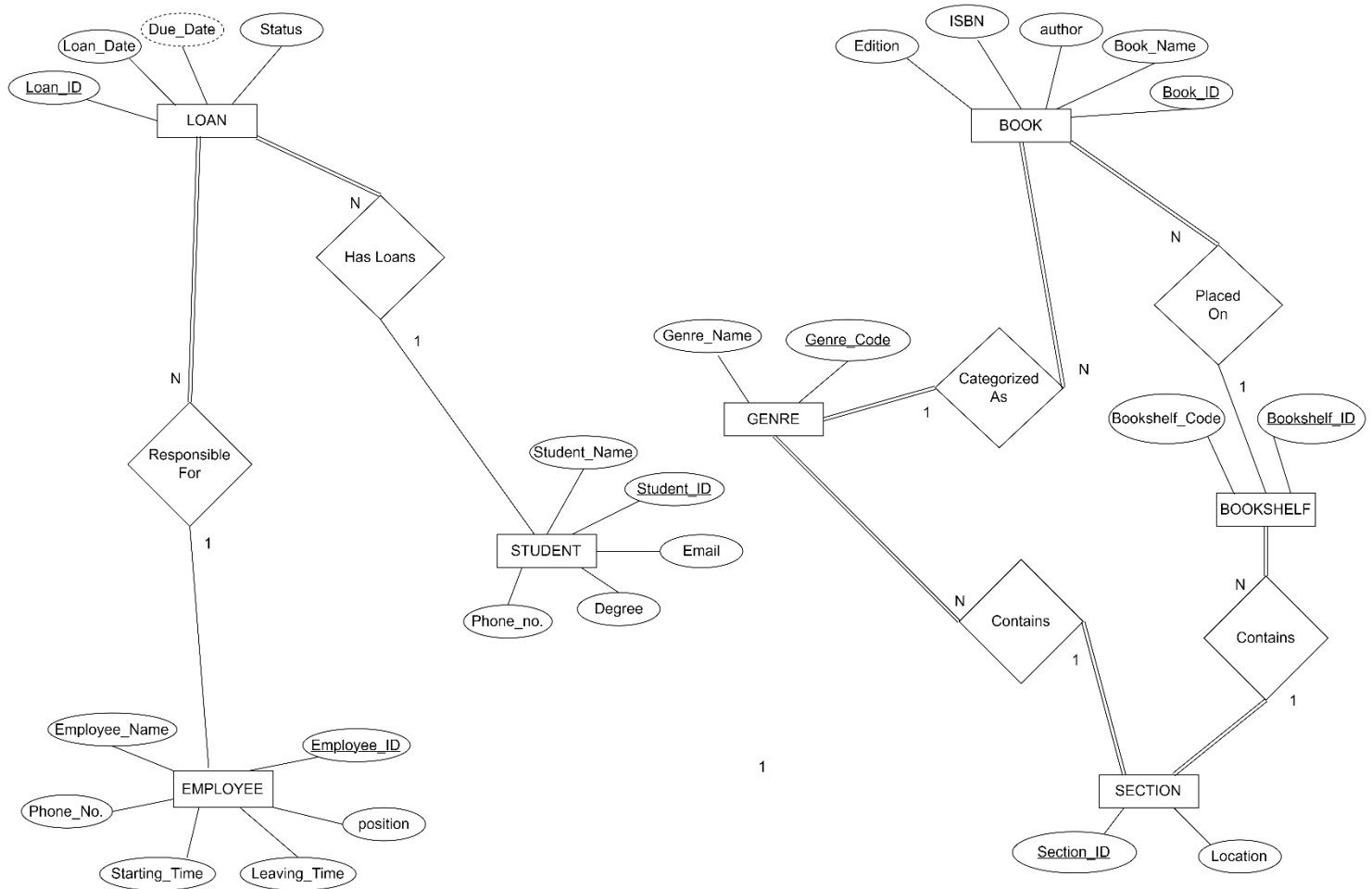
SECTION

<u>Section_ID</u>	Location	Manager_ID
-------------------	----------	------------

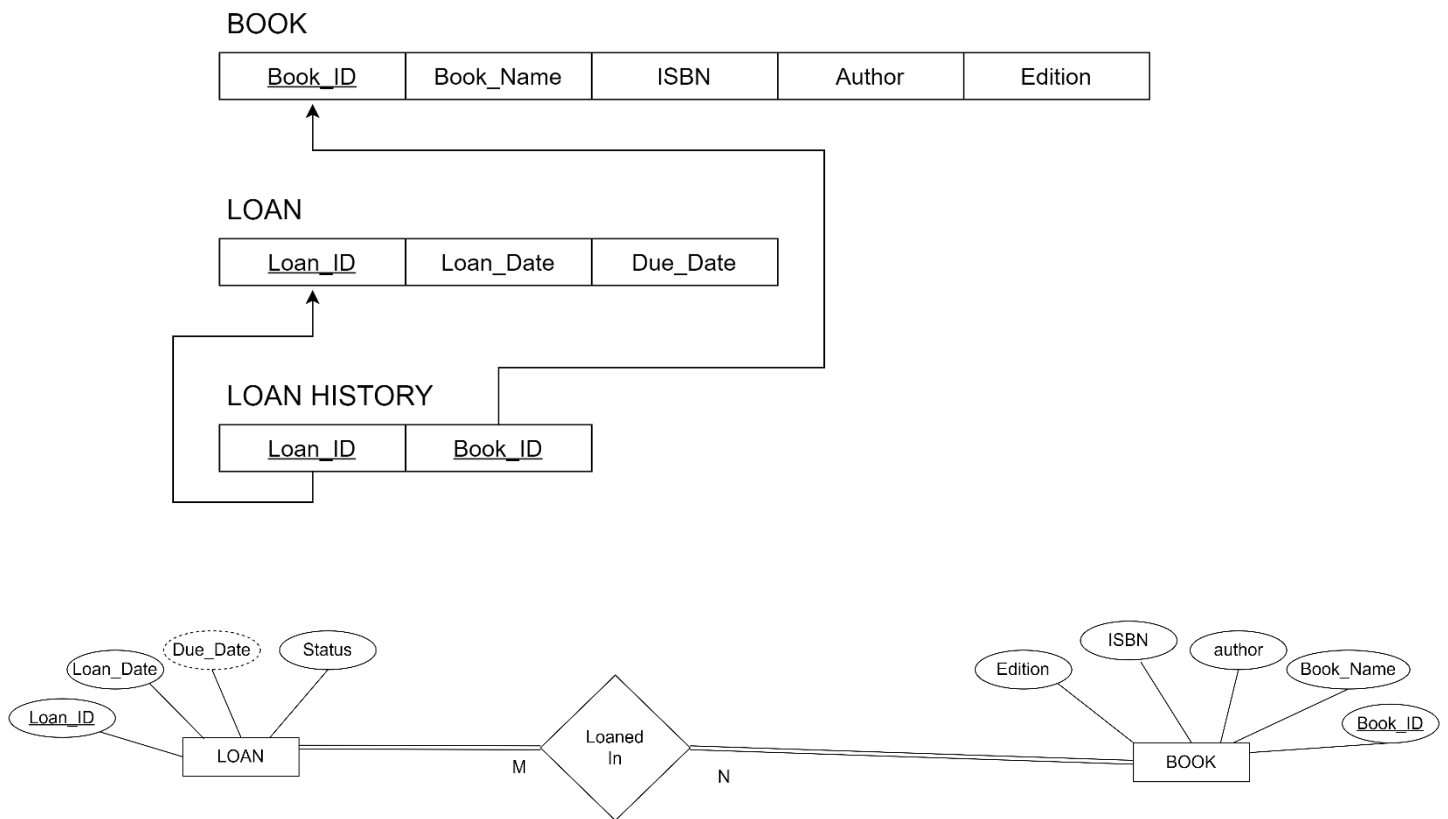


3.4 Mapping of Binary 1-N Relationship Types

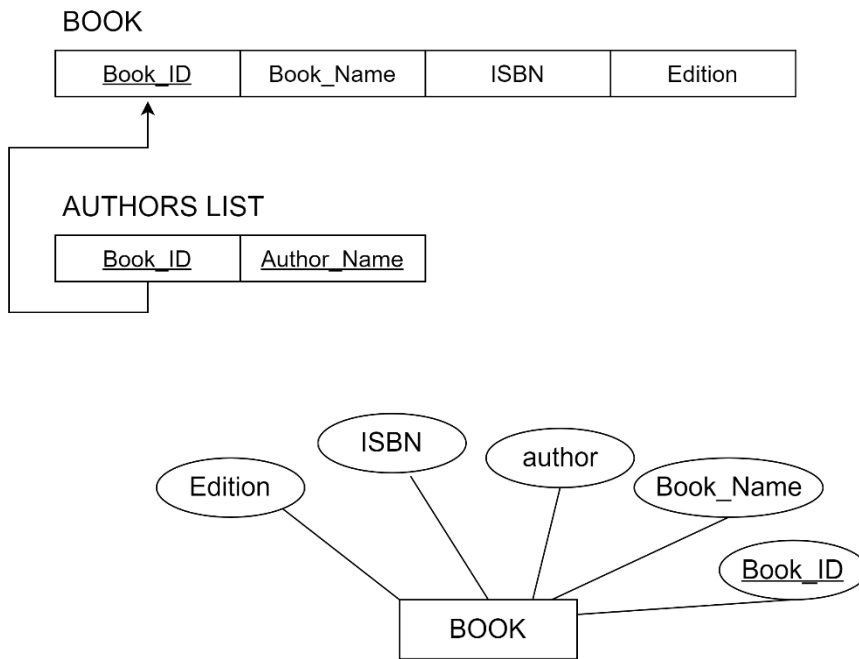




3.5 Mapping of Binary M-N Relationship Types



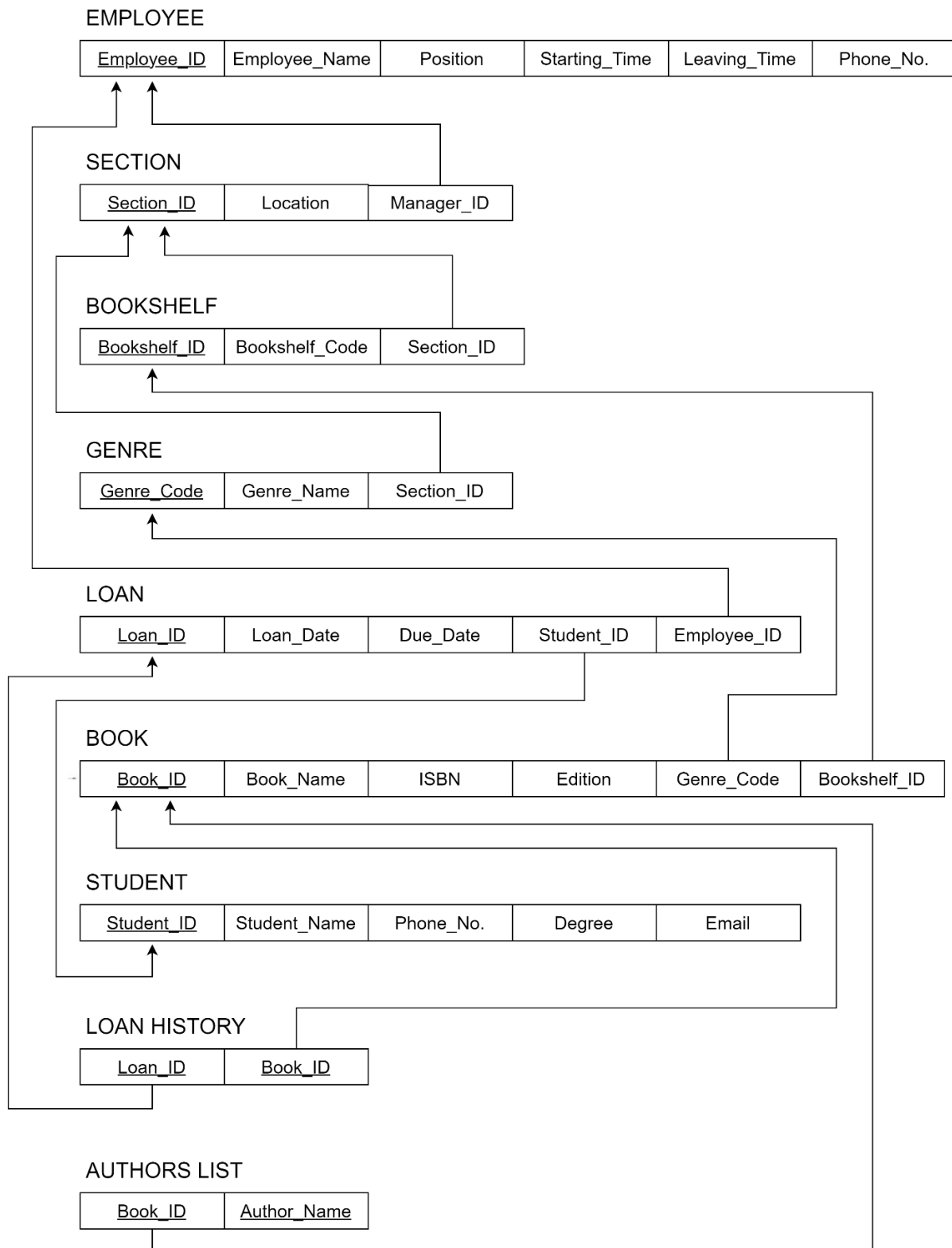
3.6 Mapping Multivalued Attributes



3.7 Mapping Multivalued Attributes

No N-ary relationship.

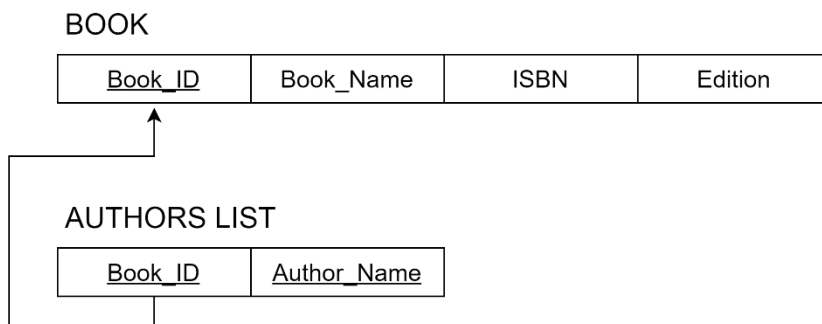
3.7 Schema Diagram



4. Normalization

4.1 First Normal Form

Decompose multivalued attributes as we did in the previous step:



The schema after 1NF:

EMPLOYEE

<u>Employee_ID</u>	Employee_Name	Position	Starting_Time	Leaving_Time	Phone_No.
--------------------	---------------	----------	---------------	--------------	-----------

SECTION

<u>Section_ID</u>	Location	Manager_ID
-------------------	----------	------------

BOOKSHELF

<u>Bookshelf_ID</u>	Bookshelf_Code	Section_ID
---------------------	----------------	------------

GENRE

<u>Genre_Code</u>	Genre_Name	Section_ID
-------------------	------------	------------

LOAN

<u>Loan_ID</u>	Loan_Date	Status	Student_ID	Employee_ID
----------------	-----------	--------	------------	-------------

BOOK

<u>Book_ID</u>	Book_Name	ISBN	Edition	Genre_Code	Bookshelf_ID
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STUDENT

<u>Student_ID</u>	Student_Name	Phone_No.	Degree	Email
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LOAN HISTORY

<u>Loan_ID</u>	<u>Book_ID</u>
----------------	----------------

AUTHORS LIST

<u>Book_ID</u>	<u>Author_Name</u>
----------------	--------------------

4.2 Second Normal Form

Seeing as the relations in our database doesn't have any composite keys, there are no partial dependency in any of the relations. Therefore, the schema does not change.

<u>Employee_ID</u>	Employee_Name	Position	Starting_Time	Leaving_Time	Phone_No.
	↑	↑	↑	↑	↑

SECTION

<u>Section_ID</u>	Location	Manager_ID
	↑	↑

BOOKSHELF

<u>Bookshelf_ID</u>	Bookshelf_Code	Section_ID
	↑	↑

GENRE

<u>Genre_Code</u>	Genre_Name	Section_ID
	↑	↑

LOAN

<u>Loan_ID</u>	Loan_Date	Status	Student_ID	Employee_ID
	↑	↑	↑	↑

BOOK

<u>Book_ID</u>	Book_Name	ISBN	Edition	Genre_Code	Bookshelf_ID
	↑	↑	↑	↑	↑

STUDENT

<u>Student_ID</u>	Student_Name	Phone_No.	Degree	Email
	↑	↑	↑	↑

LOAN HISTORY

<u>Loan_ID</u>	<u>Book_ID</u>
----------------	----------------

AUTHORS LIST

<u>Book_ID</u>	<u>Author_Name</u>
----------------	--------------------

4.3 Third Normal Form

Eliminate transitive dependencies:

Here, the Leaving_Time is dependent on the Starting_Time, so we decomposed it into two relations:

Before 3NF

EMPLOYEE

<u>Employee_ID</u>	Employee_Name	Position	Phone_No.	Starting_Time	Leaving_Time
	↑	↑	↑	↑	↑

After 3NF

EMPLOYEE

<u>Employee_ID</u>	Employee_Name	Position	Phone_No.	Starting_Time
	↑	↑	↑	↑

SHIFT

<u>Starting_Time</u>	Leaving_Time
	↑

The schema after 3NF:

EMPLOYEE

<u>Employee_ID</u>	Employee_Name	Position	Starting_Time	Phone_No.
	↑	↑	↑	↑

SECTION

<u>Section_ID</u>	Location	Manager_ID
	↑	↑

BOOKSHELF

<u>Bookshelf_ID</u>	Bookshelf_Code	Section_ID
	↑	↑

GENRE

<u>Genre_Code</u>	Genre_Name	Section_ID
	↑	↑

LOAN

<u>Loan_ID</u>	Loan_Date	Status	Student_ID	Employee_ID
	↑	↑	↑	↑

BOOK

<u>Book_ID</u>	Book_Name	ISBN	Edition	Genre_Code	Bookshelf_ID
	↑	↑	↑	↑	↑

STUDENT

<u>Student_ID</u>	Student_Name	Phone_No.	Degree	Email
	↑	↑	↑	↑

LOAN HISTORY

<u>Loan_ID</u>	<u>Book_ID</u>
----------------	----------------

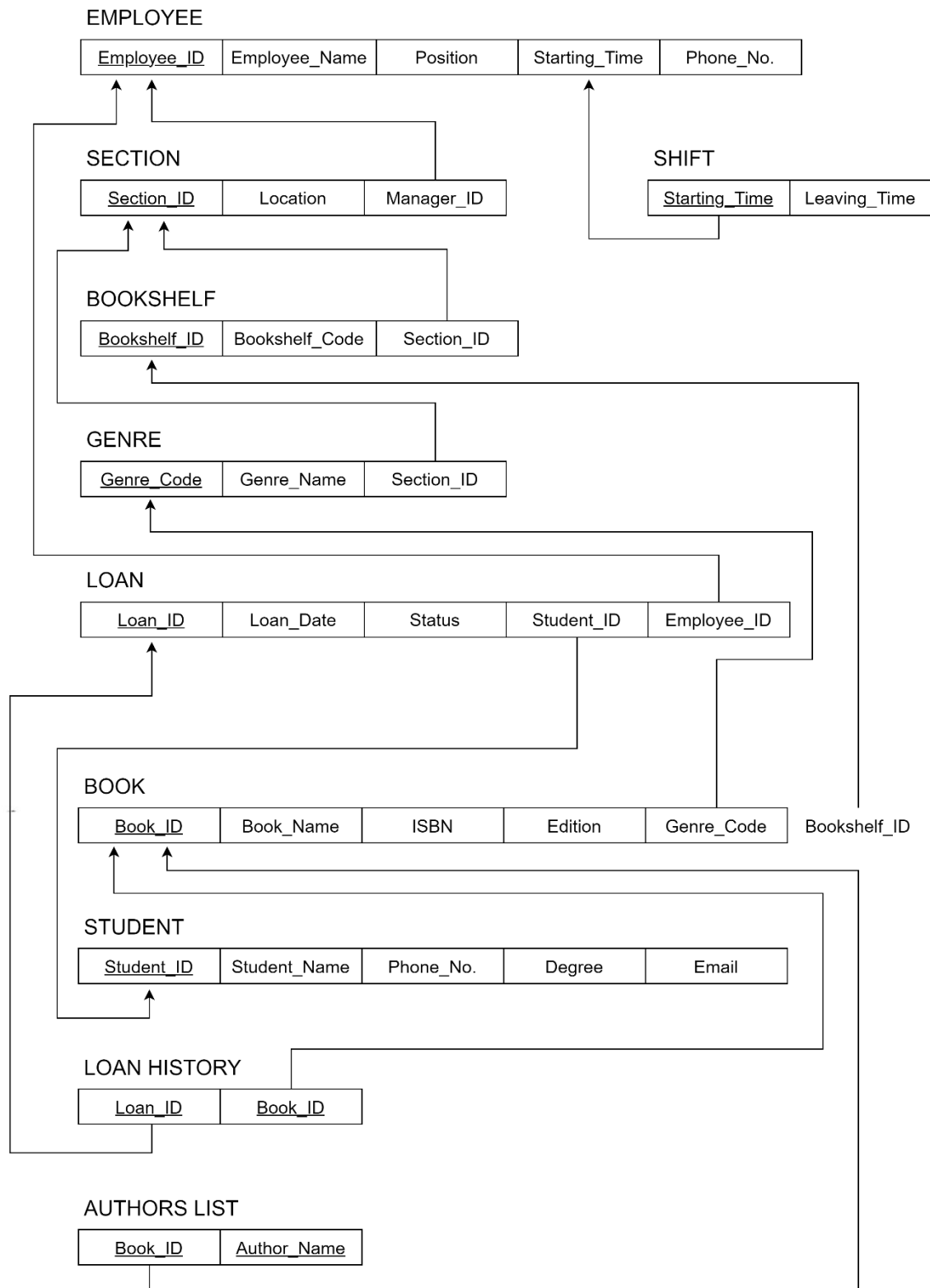
AUTHORS LIST

<u>Book_ID</u>	<u>Author_Name</u>
----------------	--------------------

SHIFT

<u>Starting_Time</u>	Leaving_Time
	↑

5. Final DB Schema Diagram



PART III: IMPLEMENTATION

6. Table Creation Script

6.1 <EMPLOYEE> Table

```
Create Table Employee (  
Employee_ID number (9,0) not null    primary key,  
Emplyee_Name varchar2 (30) not null,  
Position varchar2 (30) not null,  
Starting_Time date,  
Phone_No. number (10,0) not null,  
FOREIGN KEY (Starting_Time) REFERENCES Shifts  
(Starting_Time) ON DELETE CASCADE,  
);
```

6.2 <SHIFTS> Table

```
Create Table Shifts (  
Starting_Time date not null primary key,  
Leaving_Time date not null,  
);
```

6.3 <STUDENT> Table

```
Create Table Student (  
Student_ID number (9,0) not null primary key,  
Student_Name varchar2 (30) not null,  
Degree varchar2 (30),  
Email varchar2 (30) not null,  
Phone_No. number (10,0) not null  
);
```

6.4 <BOOK> Table

```
Create Table Book (  
Book_ID number (6,0) not null primary key,  
Book_Name varchar2 (30) not null,  
ISBN number (13,0) not null,  
Edition number (4,0) not null,  
Genre_Code number (2) not null,  
Bookshelf_ID number (4) not null,  
FOREIGN KEY (Genre_Code) REFERENCES Genre  
(Genre_Code) ON DELETE CASCADE,  
FOREIGN KEY (Bookshelf_ID) REFERENCES BookShelf  
(Bookshelf_ID) ON DELETE CASCADE  
);
```

6.5 <LOAN> Table

```
Create Table Loan (  
Loan_ID number (9,0) not null primary key,  
Loan_Date date,  
Employee_ID number (9) not null,  
Student_ID number (9) not null,  
Status varchar2 (30) not null,  
FOREIGN KEY (Employee_ID) REFERENCES Employee  
(Employee_ID) ON DELETE CASCADE,  
FOREIGN KEY (Student_ID) REFERENCES Student  
(Student_ID) ON DELETE CASCADE  
);
```

6.7 <GENRE> Table

```
Create Table Genre (  
Genre_Code number (2,0) not null primary key,  
Genre_Name varchar2 (30) not null,  
Section_ID number (2) not null,  
FOREIGN KEY (Section_ID) REFERENCES Section  
(Section_ID) ON DELETE CASCADE  
);
```

6.8 <BOOKSHELF> Table

```
Create Table BookShelf (  
  Bookshelf_ID number (4,0) not null primary key,  
  Bookshelf_Code varchar (3),  
  Section_ID number (2) not null,  
  FOREIGN KEY (Section_ID) REFERENCES Section  
  (Section_ID) ON DELETE CASCADE  
);
```

6.9 <SECTION> Table

```
Create Table Section (  
  Section_ID number (2,0) not null primary key,  
  Location varchar (15),  
  Employee_ID number (9) not null,  
  FOREIGN KEY (Employee_ID) REFERENCES Employee  
  (Employee_ID) ON DELETE CASCADE  
);
```

6.10 <LOAN HISTORY> Table

```
Create Table Loan_History (  
Loan_ID number (9) not null,  
Book_ID number (9) not null,  
CONSTRAINT PK_LoanHistory PRIMARY KEY (Loan_ID,  
Book_ID),  
FOREIGN KEY (Loan_ID) REFERENCES Loan (Loan_ID)  
ON DELETE CASCADE,  
FOREIGN KEY (Book_ID) REFERENCES Book (Book_ID)  
ON DELETE CASCADE  
);
```

6.11 <AUTHOR LIST> Table

```
Create Table Authors_List (  
Book_ID number (6) not null,  
Author_Name varchar2 (30) not null,  
CONSTRAINT PK_LoanHistory PRIMARY KEY (Loan_ID,  
Book_ID),  
FOREIGN KEY (Book_ID) REFERENCES Book (Book_ID)  
ON DELETE CASCADE  
);
```

7. Constraints Script

The referential integrity constraints, which were added to the database during its creation in section 6, are the main constraints in our database. They resulted from relations that use foreign keys to point to a primary key.

Here are some additional limitations to further highlight the database and provide examples for many kinds that we employed.

	Business Rule	SQL Script	Table
1	“Employee’s age must be between 18 years old and 60 years old”.	<pre>ALTER TABLE EMPLOYEE ADD CONSTRAINT CHK_AGE CHECK (age> 18) AND (age<60)</pre>	Employee
2	“Student’s age must be over 15 years old”.	<pre>ALTER TABLE STUDENT ADD CONSTRAIN CHK_AGE CHECK (age>15)</pre>	Student
3	“Employee ID must not be null and must be unique for each one”.	<pre>Primary key constraint was added while creating the table: create table Employee(Employee_ID number(9,0) not null primary key,...)</pre>	Employee

8. Queries and Transactions

8.1 <First Query>

Query in Natural Language (English):

Retrieve number of Books for each Author.

SQL Script:

```
SELECT BookID, BookName, AuthorName,  
COUNT(Book.BookID) AS BookCount  
FROM AuthorsList NATURAL JOIN Book  
GROUP BY AuthorsList.AuthorName;
```

The Output:

BookID	BookName	AuthorName	BookCount
012034	Language and Mind	Noam Chomsky	1
013547	Introduction to Linguistics: The Science of Language	Victoria Fromkin	1
013547	Introduction to Linguistics: The Science of Language	Robert Rodman	1
013547	Introduction to Linguistics: The Science of Language	Nina Hyams	1
021367	Meeting Muhammad	Suleiman Omar	1
023554	Disease of the heart and their cure	Ibn Taymiya	1
051357	Endurance	Alfred Lansing	1
032674	The Wealth of Nations	Adam Smith	1
031567	Thinking, Fast and Slow	Daniel Kahneman	1
043567	The Art of Computer Programming	Donald E. Knuth	1
041134	Fundamentals of Database Systems	Elmasri	1
041134	Fundamentals of Database Systems	navathe	1

8.2 <Second Query>

Query in Natural Language (English):

Find books that have been borrowed more than or equal twice.

SQL Script:

```
SELECT BookID, BookName
FROM Book
WHERE BookID IN (
    SELECT BookID
    FROM LoanHistory
    GROUP BY BookID
    HAVING COUNT(*) >= 2
    ORDER BY BookID DESC );
```

The Output:

BookID	BookName
012034	Language and Mind

8.3 <Third Query>

Query in Natural Language (English):

Retrieve the books on loan.

SQL Script:

```
SELECT BookID, BookName
FROM Book
WHERE Book.ID IN (SELECT DISTINCT
LoanHistory.BookID FROM LoanHistory);
```

The Output:

BookID	BookName
012034	Language and Mind
021367	Meeting Muhammad
023554	Disease of the heart and their cure
051357	Endurance
031567	Thinking, Fast and Slow
041134	Fundamentals of Database Systems
013547	Introduction to Linguistics: The Science of Language

8.4 <Forth Query>

Query in Natural Language (English):

List the books on specific BookShelves.

SQL Script:

```
SELECT BookID, BookName, BookShelfID
FROM Book B JOIN BookShelf S ON B.BookShelfID =
S.BookShelfID
GROUP BY BookShelfID
HAVING BookShelfID BETWEEN 101 AND 300;
```

The Output:

BookID	BookName	BookShelfID
051357	Endurance	0166
032674	The Wealth of Nation	0113
031567	Thinking, Fast and Slow	0133
043567	The Art of Computer Programming	0211
041134	Fundamentals of Database Systems	0221

8.5 <Fifth Query>

Query in Natural Language (English):

Get students who have borrowed books.

SQL Script:

```
SELECT StudentID, StudentName, PhoneNumber  
FROM Student  
WHERE StudentID IN (SELECT DISTINCT StudentID  
FROM Loan)  
ORDER BY StudentID DESC;
```

The Output:

StudentID	StudentName	PhoneNumber
2256578	Ali Alshahri	552740825
2245343	Anas Alharbi	553428597
2190876	Rami Mohammed	553704716
2123436	Mousa Fahad	552437142
2023434	Majed Alhazmi	553748122
2020543	Saeed Khaled	552410954

APPENDIX

Employee				
Employee ID	Employee Name	Position	Starting Time	Phone Number
000153692	Turki Aghamdi	Director General	8:00 AM	0556784390
000167005	Hussain Ali	Department Manager	8:00 AM	0552348720
000153772	Ahmed Alzahrani	Department Manager	8:00 AM	0556309148
000192291	Abdullah Sami	Department Manager	8:00 AM	0551739540
000136534	Saleh Alharbi	Department Manager	8:00 AM	0557847192

Shift	
Starting_Time	Leaving_Time
8:00 AM	2:00 PM
2:00 PM	9:00 PM

Student				
Student ID	Student Name	E-mail	Phone No.	Degree
2245343	Anas Alharbi	Anas02@gmail.com	0553428597	Bachelor
2256578	Ali Alshahri	Ali22@gmail.com	0552740825	Bachelor
2190876	Rami Mohammed	RM21@gmail.com	0553704716	Master
2020543	Saeed Khaled	SaeedK@gmail.com	0552410954	Bachelor
2023434	Majed Alhazmi	MA23@gmail.com	0553748122	Bachelor
2123436	Mousa Fahad	MousaF2@gmail.com	0552437142	Bachelor

Book					
Book ID	Book Name	ISBN	Edition	Genre_Code	BookShelf_ID
012034	Language and Mind	978-05216749352	3rd	01	0043
013547	Introduction to Linguistics: The Science of Language	978-13375595772	11th	01	0016
021367	Meeting Muhammad	978-1847741776	1st	02	0087
023554	Disease of the heart and their cure	978-1541348707	1st	02	0094
051357	Endurance	978-0465062881	1st	05	0166
032674	The Wealth of Nation	978-0553585971	7th	03	0113
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230524875	6-Aug-23	2245343	000167005	On Shelf
230463091	9-Aug-23	2190876	000153772	On Shelf
230586208	18-Aug-23	2020543	000167005	Over Due
230286479	27-Aug-23	2256578	000167005	On Shelf
230466759	7-Sep-23	2245343	000192291	On Shelf
230123375	11-Sep-23	2023434	000153772	On Shelf
230142533	20-Sep-23	2123436	000192291	Checked Out
230133426	27-Sep-23	2256578	000136534	Checked Out

Section		
Section ID	Section Location	Manager ID
01	First-Floor	000167005
02	First-Floor	000153772
03	Second-Floor	000192291
04	Second-Floor	000136534

Genre		
Genre Code	Genre Name	Section ID
01	Languages and Linguistics	01
02	Islamic	01
03	Business and Economics	02
04	Technology and Computer Science	03
05	History and Geography	02

Book Shelf		
Book Shelf ID	Book Shelf Code	Section ID
0001	A-001	01
0022	A-022	01
0053	A-053	01
0004	A-004	01
0198	B-098	02
0156	B-056	02
0322	D-022	04
0287	C-087	03
0290	C-090	03

Authors List	
Book ID	Author Name
012034	Noam Chomsky
013547	Victoria Fromkin
013547	Robert Rodman
013547	Nina Hyams
021367	Suleiman Omar
023554	Ibn Taymiya
051357	Alfred Lansing
032674	Adam Smith
031567	Daniel Kahneman
043567	0Donald E. Knuth
041134	Elmasri
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230524875	012034
230463091	013547
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230123375	043567
230142533	041134
230133426	012034